

Homework 0

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Problem 1

The first derivative of the function $f(x) = \exp(\sin(x))$ was computed on the interval $0 \leq x \leq 2\pi$. The numerical approximation uses second-order centered finite differences for the internal grid points and first-order one-sided differences at the boundaries.

Figure 1 shows the analytical solution plotted alongside the numerical approximation, as well as the absolute error between the two. Error is largest at the ends.

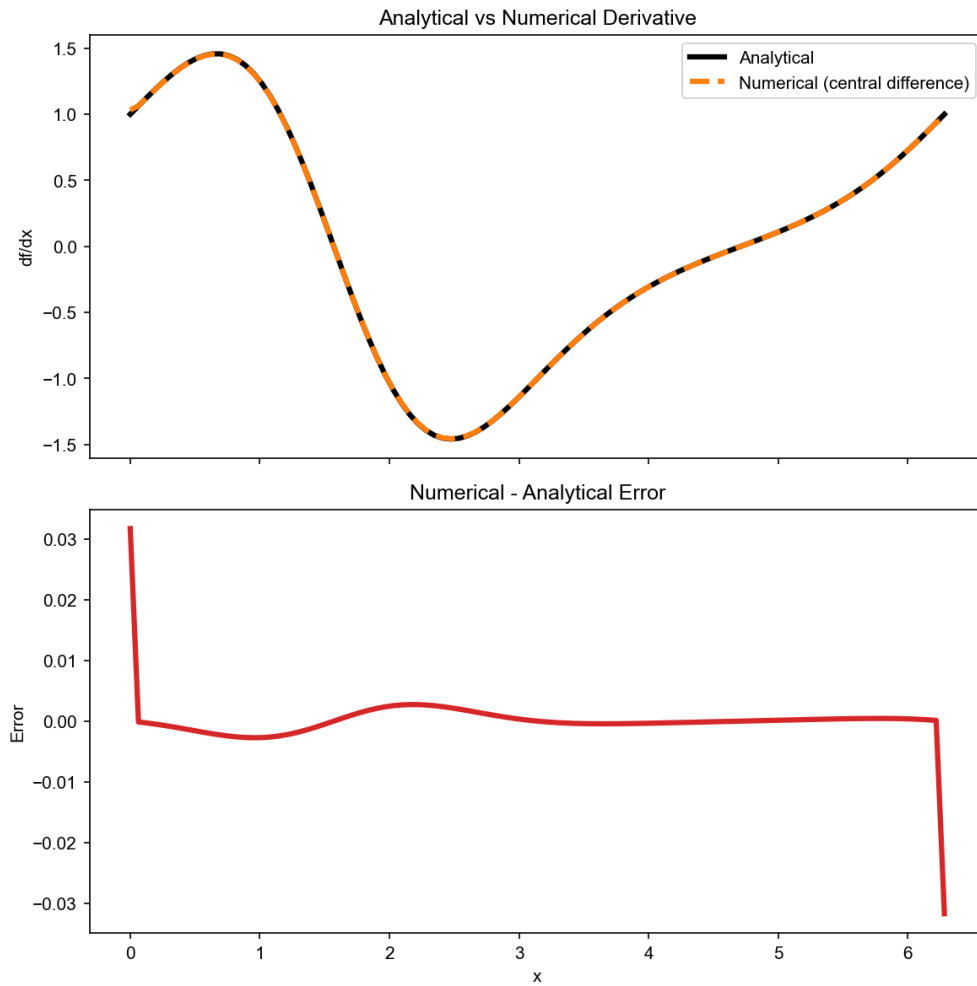


Figure 1: Comparison of analytical and numerical first derivatives of $f(x)$.

Convergence plots for a range of N values showed the relative error under the L_∞ and L_2 norms against the grid size N

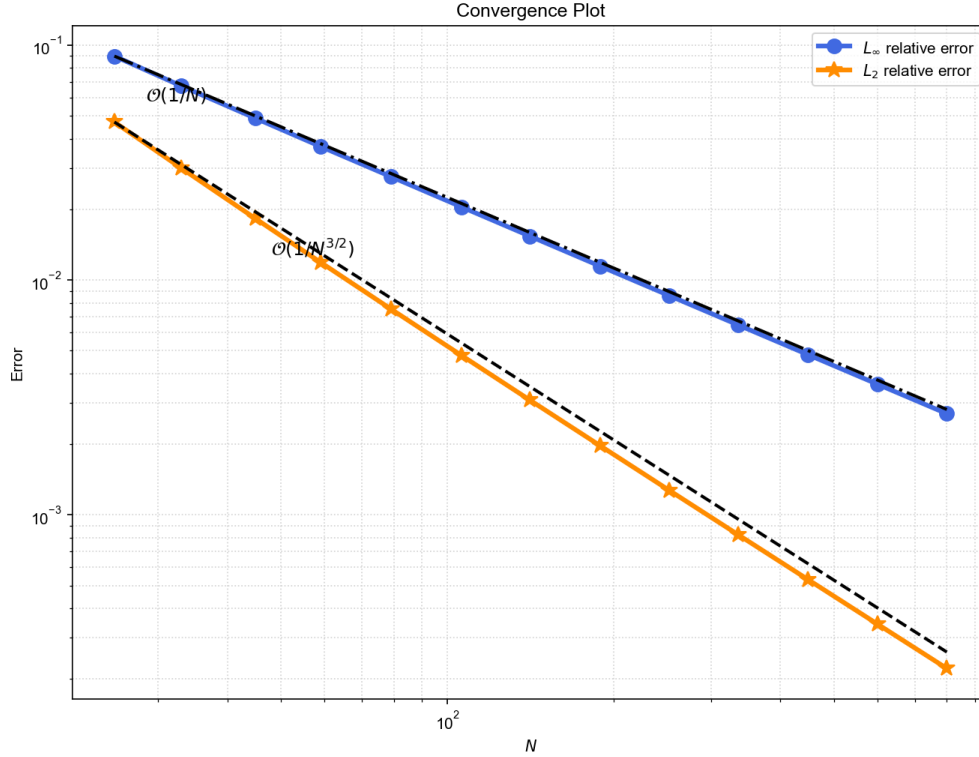


Figure 2: Relative error convergence using L_∞ and L_2 norms.

Extra Credit: The 3/2 Slope for the L_2 Norm

The convergence rates are obtained analytically by Taylor-expanding each finite-difference stencil about a grid point giving the error estimation at that point: $O(h^2)$ for the central-difference approximation used at interior nodes, and $O(h)$ for the one-sided approximation used at the two boundary nodes. L_∞ norm reports only the single largest error in the domain, it is dominated entirely by the lower-order boundary stencil resulting in a global rate of $O(1/N)$. The L_2 relative error additionally normalizes by $|f'(x)|_2$, which grows as $O(N^{1/2})$ since it aggregates an increasing number of similarly-sized terms as the grid is refined; dividing the $O(1/N)$ error norm by this growing quantity yields the compounded rate $O(N^{-3/2})$.

The theoretical predictions are verified empirically by computing the local convergence rate between successive grid refinements, $p \approx \log(e_{i+1}/e_i)/\log(N_{i+1}/N_i)$, from the simulation data. The measured rates converged to approximately -1 and $-3/2$ for the L_∞ and L_2 norms, respectively.

Problem 2

The second derivative $f''(x)$ was computed using a second-order central difference with periodic boundary conditions. Figure 3 shows the comparison of analytical and numerical second derivatives of $f(x)$

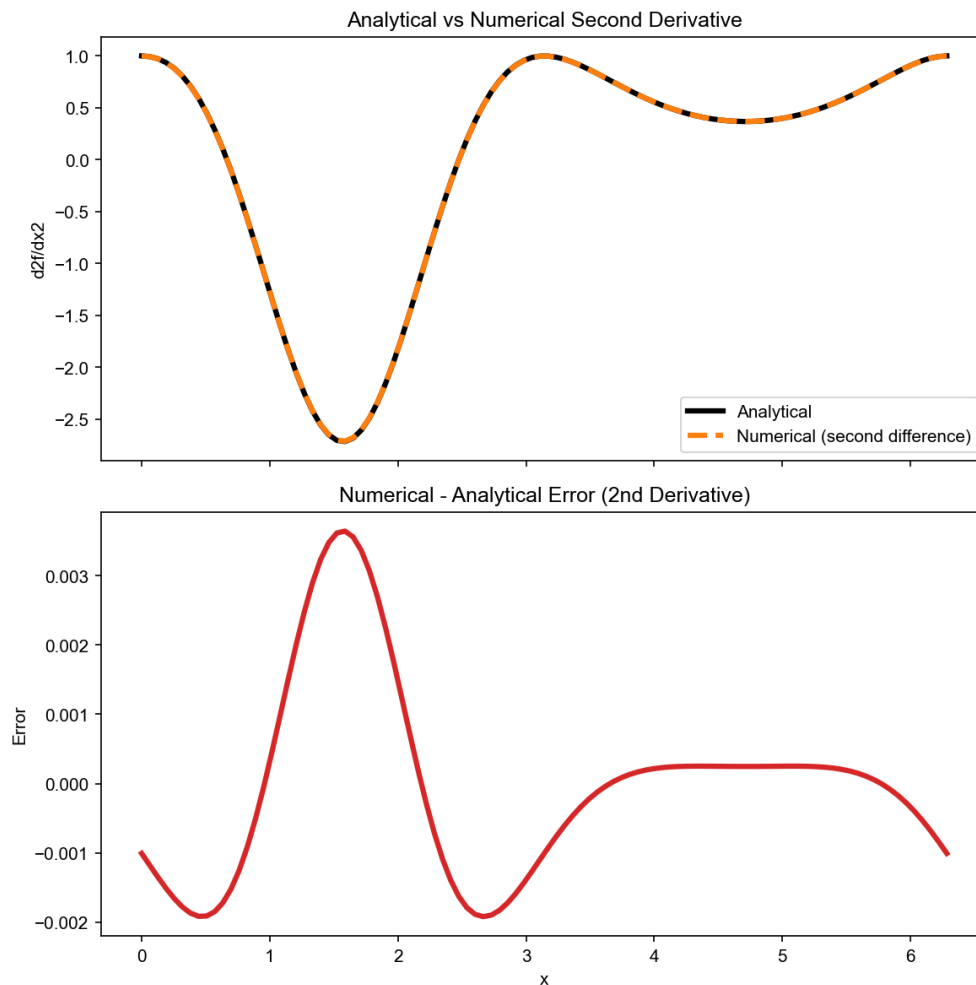


Figure 3: Comparison of analytical and numerical second derivatives of $f(x)$.

Convergence plots for a range of N values showed the relative error under the L_∞ and L_2 norms against the grid size. The error scaling confirms the expected $\mathcal{O}(1/N^2)$ second-order convergence for both norms, as the periodic boundary conditions avoids the introduction of first-order boundary errors.

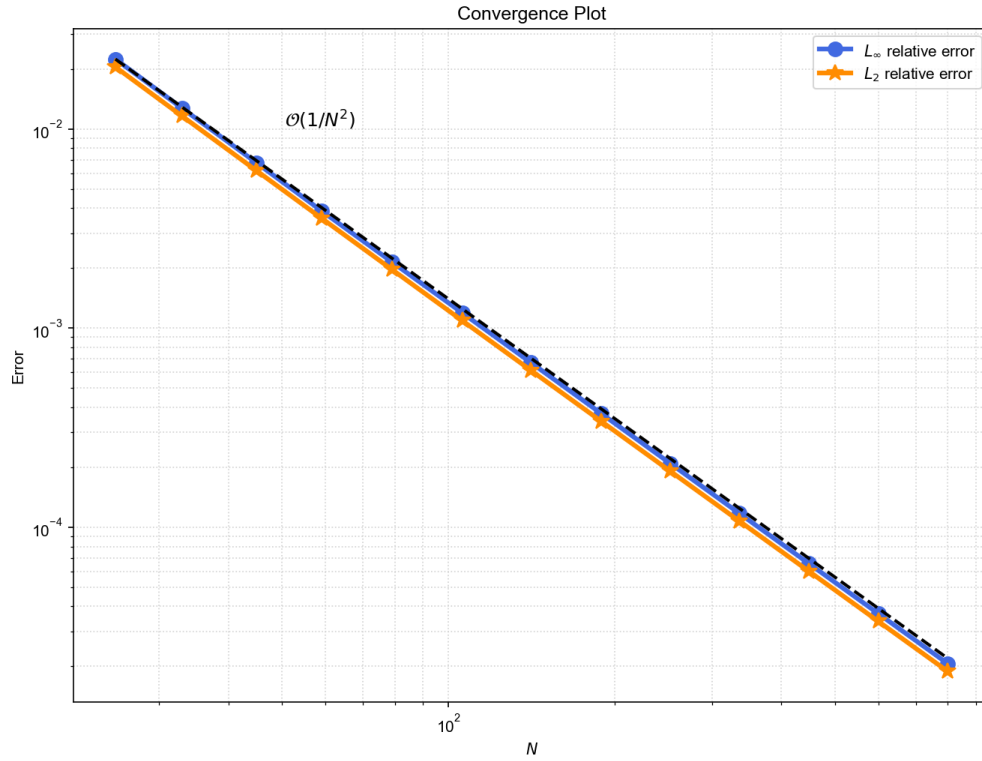


Figure 4: Relative error convergence confirming $\mathcal{O}(1/N^2)$ accuracy for the second derivative.

Problem 3

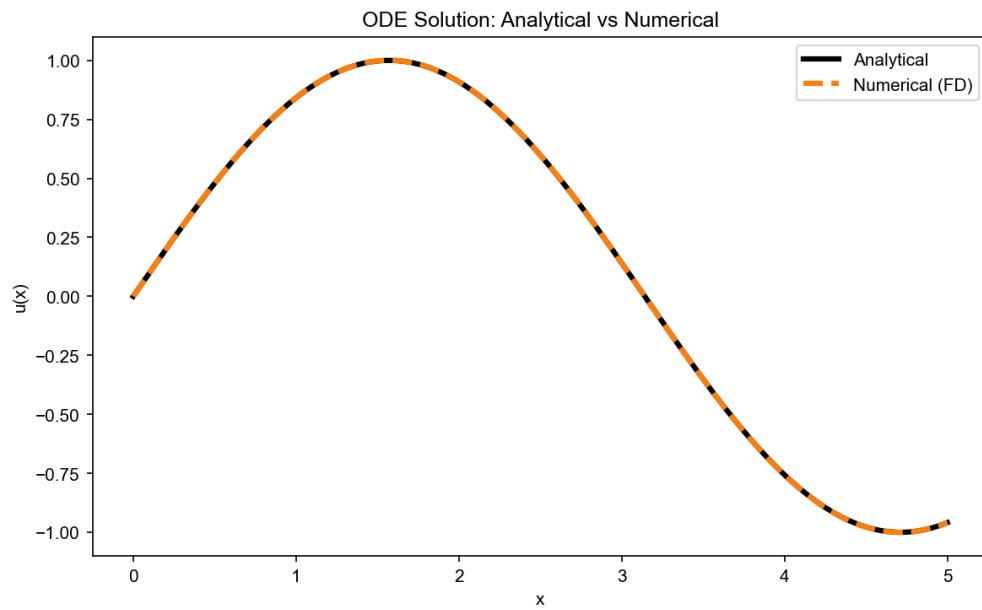


Figure 5: Comparison of analytical and numerical solutions for the ODE.

The tri-diagonal stencil was derived by substituting the second-order central-difference approximations $u'' \approx (u_{i+1} - 2u_i + u_{i-1})/h^2$ and $u' \approx (u_{i+1} - u_{i-1})/(2h)$ directly into the governing ODE, then collecting like terms in u_{i-1} , u_i , and u_{i+1} to obtain the coefficients $a_i = 1/h^2 - \sin(x_i)/2h$, $b_i = -2/h^2 + 1$, and $c_i = 1/h^2 + \sin(x_i)/2h$. The resulting finite-difference solution is a good approximation of the analytical solution $u(x) = \sin(x)$ across the domain.

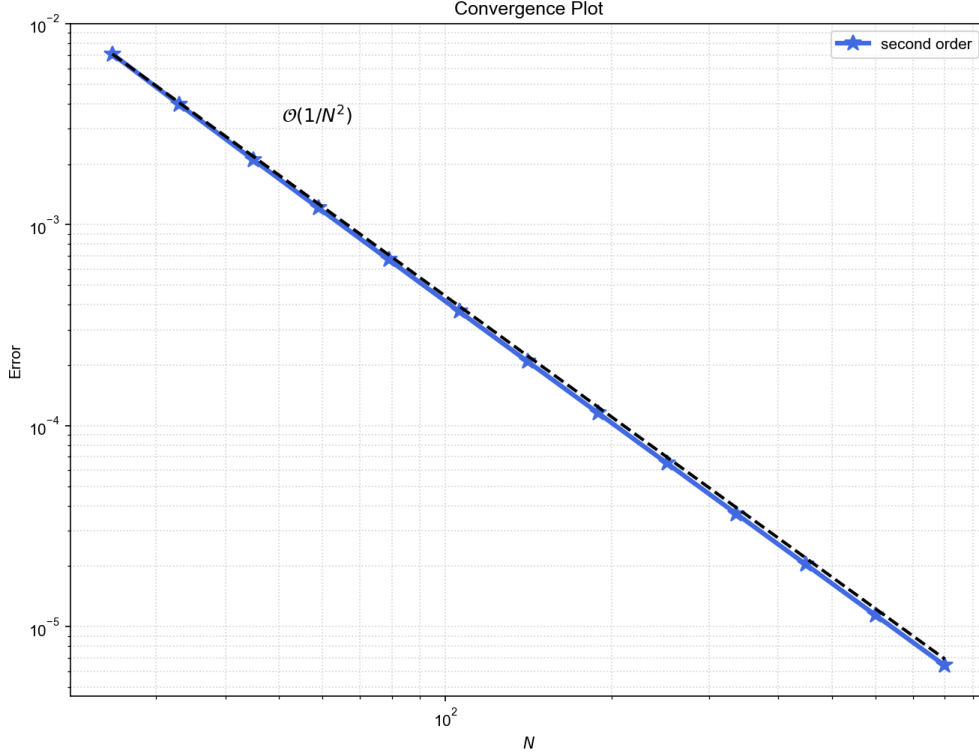


Figure 6: Relative error convergence confirming second-order accuracy.

The convergence plots for a range of N values showed the relative L_2 error against the grid size N , showing a second-order convergence $\mathcal{O}(1/N^2)$ for the finite difference scheme implemented.

Problem 4

The Poisson equation $\nabla^2 u = f(x, y)$ was solved over the two-dimensional domain $x \in [0, 5], y \in [0, 5]$ using a second-order finite difference method. Dirichlet boundary conditions were applied on all boundaries to satisfy the test solution $u(x, y) = \sin(x) \cos(y)$. The analytical derivation of the forcing function resulted in $f(x, y) = -2 \sin(x) \cos(y)$. Figure 7 displayed the analytical solution, the numerical finite difference approximation, and the numerical-analytical error distribution.

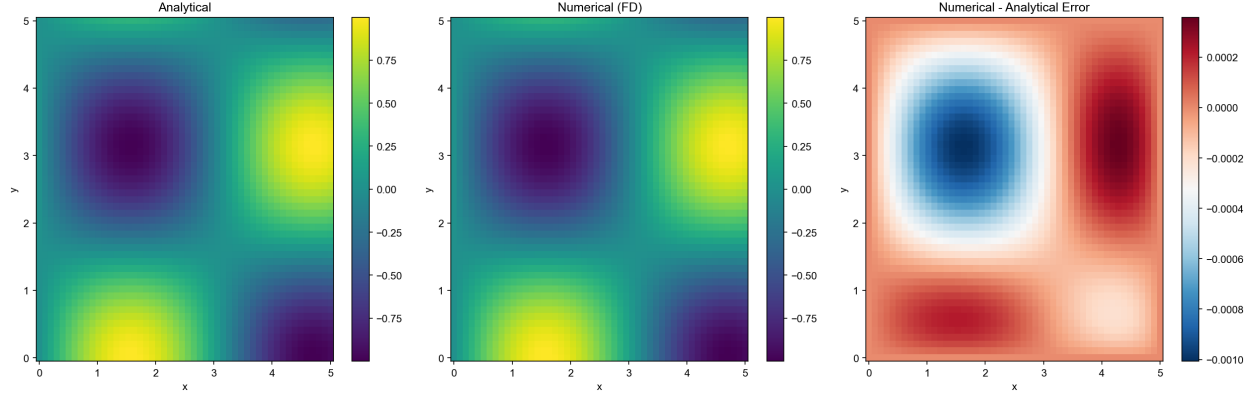


Figure 7: Comparison of analytical and numerical solutions for the Poisson equation.

The error-generating term is proportional to u itself, so it's largest in magnitude where u is largest in magnitude

Convergence plots for a range of N values showed the relative L_2 error scaling against the grid size N . The error scaling is $\mathcal{O}(1/N^2)$.

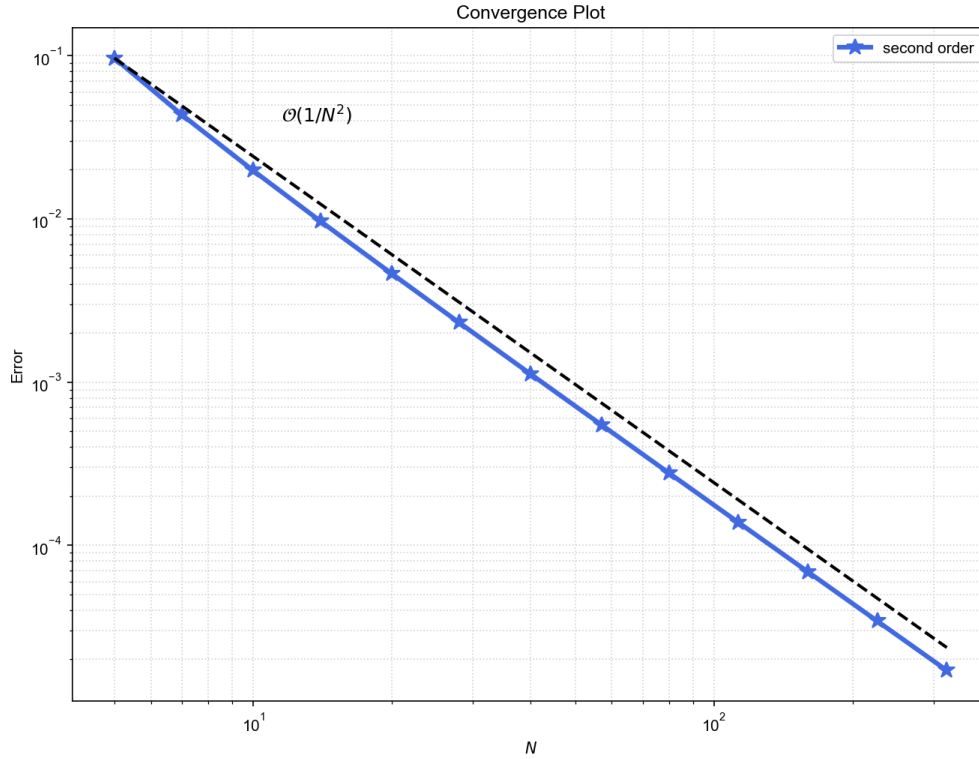


Figure 8: Relative error convergence(second-order accuracy for the Poisson equation).

Why pure Neumann conditions fail

Every row's coefficients sum to zero (interior: $1 + 1 + 1 + 1 - 4 = 0$; Neumann boundary: $\pm 1/h \mp 1/h = 0$), so the constant vector is exactly in the null space of A . The matrix is

singular, confirmed numerically by $\max |A\mathbf{1}| = 0$. The solver does not raise an error on this; it silently returns an unreasonable solution with norm $\sim 10^{15}$. This matches the PDE itself: a pure-Neumann Poisson problem requires

$$\int_{\Omega} f \, dA = \oint_{\partial\Omega} \frac{\partial u}{\partial n} \, ds \quad (1)$$

to have a solution at all, and even then u is only defined up to an arbitrary additive constant, exactly the null direction found above.

Fix: pin u at one grid point (Dirichlet) to remove the null space. This only works if the pinned point is interior (or a non-corner boundary point), pinning a corner leaves the matrix rank-deficient by 1, as verified directly.

This explanation was developed with the assistance of Claude (Anthropic) Model: Sonnet(High).

Acknowledgment

All plotting routines used to generate the figures in this report were developed with the assistance of Claude (Sonnet 5), an AI model developed by Anthropic.